

Your grade: 100%

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Next item →

1. Which of the following is NOT a valid application of a rule of inference in predicate logic? 1 / 1 point
- ☐

From $P(c)$, conclude $\exists x P(x)$

☒

From $P(c)$, conclude $\forall x P(x)$

☐

From $\exists x P(x)$, conclude $P(c)$ for an instance c

☐

From $\forall x(P(x) \rightarrow Q(x))$, conclude $P(c) \rightarrow Q(c)$

Nice work
This is not obviously true, because knowing $P(c)$ does not guarantee that P is correct for every instance.

2. To prove the statement "if n is odd, then n^2 is odd" by contrapositive, which statement should be proven? 1 / 1 point
- ☐

if n^2 is even, then n is even.

☐

if n^2 is odd, then n is even.

☐

if n is even, then n^2 is even.

☐

if n^2 is even, then n is odd.

Nice work

3. Which of the following is the contrapositive of if $x^2 > 16$, then $x > 4$? 1 / 1 point
- ☐

if $x^2 \leq 16$, then $x \leq 4$ or $x \geq -4$

☐

if $x > 4$, then $x^2 > 16$

☒

if $x \leq 4$, then $x^2 \leq 16$

☐

if $x^2 \leq 16$, then $x > 4$

Nice work

4. Which of the following is a valid conclusion from the premises $\forall x(P(x) \rightarrow Q(x))$ and $P(a)$? 1 / 1 point
- ☐

$\forall x Q(x)$

☐

$\exists y \wedge Q(y)$

☒

$Q(a)$

☐

$\neg P(a)$

Nice work
Because of the universal instantiation on $\forall x(P(x) \rightarrow Q(x))$ gives $P(a) \rightarrow Q(a)$, and modus ponens with $P(a)$ concludes $Q(a)$.

5. Which of the following is NOT a valid rule of inference? 1 / 1 point
- ☐

$\{(p \vee q), (\neg p)\} \vdash q$

☒

$\{(p \rightarrow q), q\} \vdash p$

☐

$\{p \wedge q\} \vdash p$

☐

$\{(p \rightarrow q), (\neg q)\} \vdash \neg p$

Nice work

6. Which of the following conclusions can be drawn from the premises $(p \rightarrow q)$, $(q \rightarrow r)$, p ? 1 / 1 point
- ☐

$q \wedge r$

☐

$\neg r$

☒

r

☐

$\neg p$

Nice work
Using modus ponens, $p \rightarrow q$ and p imply q , and then $q \rightarrow r$ and q imply r .

7. Which of the following is the correct contrapositive of the statement: $p \rightarrow q$? 1 / 1 point
- ☐

$q \rightarrow p$

☐

$\neg p \rightarrow \neg q$

☐

$\neg q \rightarrow p$

☒

$\neg q \rightarrow \neg p$

Nice work
The contrapositive of a conditional statement $p \rightarrow q$ is $\neg q \rightarrow \neg p$

8. Which rule of inference allows us to conclude $P(c)$ from $\forall x P(x)$ for some specific constant c ? 1 / 1 point
- ☐

Universal generalization

☐

Existential generalization

☒

Universal instantiation

☐

Existential instantiation

Nice work

9. Which of the following is NOT a valid rule of inference? 1 / 1 point
- ☒

$p \rightarrow q, \neg p \vdash \neg q$

☐

$p \rightarrow q, p \vdash q$

☐

$p \wedge q \vdash p$

☐

$p \vee q, \neg p \vdash q$

Nice work
Concluding $\neg q$ from $p \rightarrow q$ and $\neg p$ is the fallacy of denying the antecedent, which is not valid. The other options are valid rules.

10. In a proof by contradiction to show that there is no largest even integer, what is assumed? 1 / 1 point
- ☐

All even integers are finite

☐

There is no even integer

☐

There is a largest even integer

☒

There is a largest even integer, and we derive a contradiction

Nice work
We simply assume there is a largest even integer and derives a contradiction (e.g., by showing $n + 2$ is a larger even integer).